W-7. Conservation Pond/Reservoir Improvement

Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures
Water Resources Development	Medium	High		L Term

Descriptions

Measures	Reservoirs and conservation ponds are essential to preserving regional ecosystems, improving biodiversity, and managing water resources. They offer plenty of plant and animal species habitats and are essential supplies of water for drinking, agriculture, and enjoyment. In order to aid local communities, improve biodiversity, and promote sustainable water management, it is essential that conservation ponds and reservoirs be improved through efficient conservation methods. Bangladesh's CHT, deal with particular difficulties with regard to resource management and water scarcity. Because of its unique topography—which includes steep hills and erratic rainfall—the area requires efficient water saving measures. In order to collect and store rainwater, conservation ponds and reservoirs can be crucial for supplying essential water supplies for local ecosystems, households, and agriculture. The CHT' environment and inhabitants will have a sustainable future if conservation is given top priority in the management of these essential resources. This will ensure a balanced approach to ecological preservation and water resource development.

Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■

Applicable Sites	• Moderate to Steep Slopes (5-15%) – Requires terracing and reinforced embankments. • Low-Lying Catchment Areas – Naturally collects runoff, reducing excavation costs. Deforested or Degraded Land – Helps in soil conservation and groundwater recharge. • Near Agricultural Fields & Villages – Provides easy access for irrigation and household use.

Slope	Less than 3%	3% to 15%	15% to 30%	30% to 60%	Above 60%
	√	√			

Design options and performance	• Suitable for low-lying areas, allowing natural runoff collection. • Constructed on slopes with reinforced bunds to store water. • Step-like water storage structures ideal for hilly terrain, reducing soil erosion. • Designed for household, livestock, and irrigation use with filtering mechanisms.
Feasibility criteria	**Technical Feasibility** • Ensures reduced seepage and minimal erosion.

<table>
<tr><td></td><td>• Assesses catchment size and runoff potential.
• Ensures adequate depth, storage volume, and outlet structures.
Environmental Feasibility
• Prevents runoff-related degradation.
• Enhances groundwater replenishment.
• Supports local flora and fauna by creating aquatic habitats.
Economic Feasibility
• Utilizes locally available materials for excavation and lining.
• Improves crop yields, food security, and household water availability.
• Encourages community-led management to reduce financial burden.
Social Feasibility
• Engages local populations in construction and management.
• Supports farming, fisheries, and household water supply.</td></tr>
<tr><td>Operation and maintenance</td><td>• Routine Maintenance – embankment repair, and vegetation control.
• Structural Repairs – Ensures bund stability, spillway function, and leak prevention.
• Seasonal Adjustments – Prepares for monsoon overflow and dry-season water conservation
• Community Involvement – Establishes local water user groups for monitoring and upkeep.</td></tr>
<tr><td>Cost</td><td>Medium</td></tr>
<tr><td>Benefits</td><td>Benefits are as follows-</td></tr>
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Type	**Low**	**Medium**	**High**
Canopy Cover Increase			
Carbon Sequestration			
Crop Production Increase			
Crop Failure Risk Reduction			
Ensuring Food Security			
Fruit Production Increase			
Forestry Goods Production Increase			
Livestock Production Increase			
Fisheries Production Increase			
Alternative Livelihood Generation			
Increase of Income			
Soil Erosion Control			
Improved Soil Health			
Soil Fertility Increase			
Sustainable Water Supply			
Water Quality Improvement			
Water Purification			
River Bank Erosion Control			
Surface Runoff Reduction			

	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	**Low**	**Medium**	**High**
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• **Water Conservation** – Enhances groundwater recharge and reduces wastage. • **Climate Resilience** – Mitigates drought effects and regulates seasonal water availability. • **Ecosystem Protection** – Supports wetland restoration and biodiversity conservation.			
Remarks	Selecting the right site is critical, as locations must be geologically stable and hydrologically viable to ensure long-term functionality. The sustainability of these projects depends on regular maintenance, with active community participation playing a key role in their success. Additionally, reservoirs should be integrated into broader watershed management efforts to enhance overall catchment conservation. Collaboration between the government and NGOs is essential, providing policy support, funding, and technical expertise to ensure efficient implementation and long-term viability.			